

Christina (Molin) Zhang

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Education

University of Toronto

Sep 2022 - Apr 2027(expected)

Bachelor of Applied Science, Computer Engineering + PEY

Relevant Courses: Algorithms and Data Structures, Software Communication & Design, Programming Fundamentals(C++), Computer Fundamental(C), Intro to Computer Programming(Python), Applied Fundamentals of Deep Learning, Computer Organization, Digital Systems, Digital Electronics, Computer networks

Skills

Software: Python, C++, C, Git, Github, QT, Linux, EZGL, GTK, API integration, VNC, VSCode, debugging, Matlab, PyTorch, Google Collab, LDRA Testbed, Network Protocols: UDP, TCP

Hardware: Assembly, Verilog, FPGA, ModelSim, Quartus Prime, LTSpice, Multisim, CPUlator, Simulink

Other: Overleaf, Nomad, Procreate, Microsoft Office, Excel, Adobe After Effects, teamwork, communication

Languages: English, Mandarin Chinese

Technical Projects

Projects can be found at [puddin1102 \(Christina Zhang\) \(github.com\)](https://github.com/puddin1102)

Team Member, GIS Application Project, U of T

Jan 2024 - Apr 2024

Skills: C++, git, Glade, EZGL, GTK, API, debugging ...

- Worked with 2 teammates to develop a mapping software in C++, using the OpenStreetMap database and API to visualize geographic information such as public transportation, schools, hospitals...
- Designed and developed the front-end user interface to allow for interactivity of map to move, zoom, or search for streets and intersections using GTK and the EZGL package
- Participated in implementing Dijkstra and A* algorithms to facilitate finding the optimal path between two user-selected points
- Utilized greedy heuristics, parallel programming, and local perturbations to solve the traveling salesman problem
- Wrote test cases to ensure software functionality, resulting in the team's code surpassing 86% of all groups in terms of stability and legality
- Collaboratively coded with teammates using GIT

Team Member, Dog Breed Classification Neural Network, U of T

May 2024 - Aug 2024

Skills: Python, PyTorch, Google Collab, Transfer Learning, Single Shot Detection(SSD)

Presentation on this project can be found at: <https://youtu.be/7yoiMFDyPE0>

- Used Google Collab to develop a breed-classifying convolutional neural network in Python with 2 friends
- Created the training, testing, and validation datasets and implemented a simple CNN baseline model to assess performance
- Used AlexNet-based transfer learning method and SSD to enhance model performance, reaching a final test accuracy of 85.69%

Pizza-Making Game, U of T

Mar 2024 - Apr 2024

Skills: C, Assembly, FPGA, CPUlator, Github, Procreate, Quartus Prime

A gameplay video can be found at: <https://www.youtube.com/watch?v=susvP2mGgVw>

- Designed and implemented a pizza-making game to work with a Nios II De1-SoC board with a partner
- Implemented for all drawing functions, VGA display, PS2 Keyboard input in C and shared code through Github
- Drew all visual components of the game by hand using Procreate
- Simulated all game code on CPUlator and utilized Quartus Prime to connect functions with the FPGA board.

Work Experience

Embedded Software Engineering Intern, CAS Space

Oct 2025 - Jan 2026

Beijing, China

- Participated in the development and testing of spacecraft software, coding primarily in C++ and QT.
- Performed unit and integration tests using LDRA Testbed and TBrn to ensure software reliability.
- Documented testing procedures, results, and software modifications for compliance and traceability

Research Assistant, Tsinghua University

Summer 2023

Beijing, China

- Translated scholarly papers for academic and research purposes
- Acquired knowledge in CubeSat robotics and gained hands-on experience in their development.
- Collaborated with a team of five members to win a national competition award

Awards

2nd Place in China

2023

The 2nd Space Universities CubeSat Challenge (SUCC)